



C:\Users\Lenovo\k_fold_cross_validation.py

py X svr.py X flask01.py X L1_L2_Reg.py X lgistic_reg.py X naive_bayes.py X decision_tree_classification.py X k_fold_cross_validation.py X grid_search.py X

```

1 # k-Fold Cross Validation
2
3 # Importing the libraries
4 import numpy as np
5 import matplotlib.pyplot as plt
6 import pandas as pd
7
8 # Importing the dataset
9 dataset = pd.read_csv(r"C:\Users\Lenovo\OneDrive\Desktop\Naresh i Note\sept\12th-09\10th - Cross validation\10th - Cross validation\1.K-FOLD CROSS VALIDATION CODE_
10 X = dataset.iloc[:, [2, 3]].values
11 y = dataset.iloc[:, -1].values
12
13 # Feature Scaling
14 from sklearn.preprocessing import StandardScaler
15 sc = StandardScaler()
16 X = sc.fit_transform(X)
17
18 # Splitting the dataset into the Training set and Test set
19 from sklearn.model_selection import train_test_split
20 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
21
22 # Training the Kernel SVM model on the Training set
23 from sklearn.svm import SVC
24 classifier = SVC(kernel = 'rbf', random_state = 0)
25 classifier.fit(X_train, y_train)
26
27 # Predicting the Test set results
28 y_pred = classifier.predict(X_test)
29
30 # Making the Confusion Matrix
31 from sklearn.metrics import confusion_matrix
32 cm = confusion_matrix(y_test, y_pred)
33 print(cm)
34
35 from sklearn.metrics import accuracy_score
36 ac = accuracy_score(y_test, y_pred)
37 print(ac)
38
39 bias = classifier.score(X_train, y_train)
40 bias
41
42 variance = classifier.score(X_test, y_test)
43 variance
44
45 # you can add implement auc & roc
46
47 # Applying k-Fold Cross Validation
48 from sklearn.model_selection import cross_val_score
49 accuracies = cross_val_score(estimator = classifier, X = X_train, y = y_train, cv = 10)
50 print("Accuracy: {:.2f} %".format(accuracies.mean()*100))
51 #print("Standard Deviation: {:.2f} %".format(accuracies.std()*100))
52
53 # -----

```

C:\Users\Lenovo

Name	Type	Size	Value
ac	float	1	0.93
accuracies	Array of float64	(10,)	[0.8 0. ...
best_accuracy	float64	1	0.9099999999
best_parameters	dict	3	{'C':1, 'gam
bias	float	1	0.9066666666
classifier	svm.classes.SVC	1	SVC object o
cm	Array of int64	(2, 2)	[[64 4] [3 29]]
dataset	DataFrame	(400, 5)	Column names

Help Variable Explorer Plots Files Code Analysis

Console 1/A X

value-mapping will have precedence in case its length matches with *x* & *y*. Please use the *color* keyword-argument or provide a 2D array with a single row if you intend to specify the same RGB or RGBA value for all points.

```
plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1],
```

In [6]: runfile('C:/Users/Lenovo/k_fold_cross_validation.py',
wdir='C:/Users/Lenovo')
[[64 4]
[3 29]]
0.93
Accuracy: 90.00 %

In [7]:

IPython Console History